Index

3DES	3:62, 3:65
6LoWPAN	3:10, 3:85-86
802.3/Ethernet	3:15

A

A5/1	3:64
Aardvark	2:47-48
ABB	1:7, 1:20, 1:22, 3:84, 3:88, 4:84
Access Control Lists (ACLs)	2:26, 2:30, 2:85, 3:7, 3:31, 3:47, 4:73
ACK	3:18
Active Directory	2:27, 2:30, 2:37, 3:89, 4:46, 4:52, 4:58, 4:62, 4:64-65, 4:71, 5:48
Advanced Encryption Standard (AES)	3:60, 3:62-63, 3:65, 3:68, 3:80, 3:84-87, 3:89
Advanced Security Acceleration Project for the Smart Grid (ASAP-SG)	1:2, 2:1, 3:1, 4:1, 5:1
Agile Software Development	5:55
AirPort	1:23
alarm	1:36, 1:43-44, 1:47, 1:50-52, 1:67, 1:75, 2:30-31, 2:86, 2:88, 3:23, 3:29, 3:56, 4:25, 5:86-87, 5:97
American National Standards Institute (ANSI)	2:65, 2:84
Analog I/O	1:36
Annualized Loss Expectancy (ALE)	5:85
Anomalies and Events (DE.AE)	5:58, 5:61
AntiVirus (AV)	3:47, 3:54, 4:25, 4:54-55, 5:32-34, 5:88
application layer	2:61, 2:63, 2:67, 2:85, 3:10, 3:18, 3:35, 3:45, 3:47, 3:53, 3:86, 3:93, 3:95, 3:98, 3:101
Application Layer Firewall	3:45, 3:47
AppLocker	4:52, 4:62, 5:34-35

ARM	1:23, 1:28, 1:30-31, 1:36, 1:43-44, 1:47, 1:50-52, 1:54, 1:67, 1:75, 2:7, 2:9, 2:30-31, 2:86, 2:88, 3:23, 3:29, 3:56, 3:79, 3:98, 4:24-25, 4:53, 4:72, 4:85, 5:35, 5:8687, 5:97-98
ARP Spoofing	3:38, 3:45
Asset Management (ID.AM)	2:73, 5:58-59
asymmetric algorithm	3:62, 3:70
attack model	2:6
Attack surfaces	2:2, 2:6, 2:8, 2:14-15, 2:22, 2:42, 2:49
AUDITPOL.EXE	5:41
Authentication Bypass	4:35-36
Authentication Header (AH)	3:12
Awareness and Training (PR.AT)	5:58, 5:60
Attack Tree	2:9
Antivirus	5:33
Application Control at Runtime	5:34
Application Control at Execution	5:35
Antivirus	5:33
Anti Spoofing	2:48

B

Background Intelligent Transfer Service (BITS)	4:58
BACnet	1:51, 3:23, 3:86
Base Transceiver Station (BTS)	3:78
Basics of Controller Logic	1:30
Bastille	5:20
batch	1:14, 1:16, 1:28-29, 1:54, 2:63, 3:26, 4:72-73
Binwalk	2:53

Biometric	1:75, 4:45, 4:52
Bitdefender	5:32
BitLocker	4:52-53
BlackEnergy2 (BE2)	4:20
Blowfish	3:62-63
Bluetooth	3:64, 3:77, 3:82, 3:84, 3:87, 3:90, 3:93-95, 3:99
Broadband Global Area Network (BGAN)	3:77, 3:79, 3:81
Buffer Overflow	2:51, 3:39, 3:45
Building Management System (BMS)	1:51
Bus Pirate	2:47-48
Business Continuity (BC)	1:8, 5:74-75, 5:88-91, 5:95
Business Environment (ID.BE)	5:58-59
Business Impact Analysis (BIA)	5:75, 5:89-90
Backups	4:88
BGAN	3:79

C

C-Band (4-8GHz)	3:79-80
Carrier Sense Multiple Access/Collision Detection (CSMA/CD)	3:5
CDMA	3:77-78
Cellular backhaul	3:78
centralized logging	5:48
chkconfig	5:18
chown	5:13
chroot()	5:15
CIP-002	5:76
CIP-009	5:76
CIS CSC	2:4, 2:41, 2:56, 3:4, 3:34, 3:44, 3:59, 3:76, 3:92, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:68, 5:82, 5:93
Cisco	2:5, 3:5, 3:11, 5:21, 5:32

Closed-Circuit TeleVision (CCTV) 1:51, 1:75, 5:87
cmdlet 4:69, 4:71, 4:75
COBIT 5 2:4, 2:41, 2:56, 2:77, 3:4, 3:34, 3:44, 3:59, 3:76, 3:92, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:68, 5:82, 5:93
Common Industrial Protocol (CIP) 1:22, 1:24, 2:63-64, 2:85, 3:26, 5:73, 5:76, 5:80
Common Vulnerability Scoring System (CVSS) 5:83
Common Weakness Enumeration (CWE) 2:12
Communication Robustness Testing (CRT) 2:84
Communications of the ACM (CACM) 3:69
Comodo 5:32
Compact Flash (CF) 2:42
Containment 5:94, 5:98-99, 5:109
Contingency Analysis 1:52, 1:55, 2:30
continuous 1:14, 1:17-18, 1:28, 1:34, 1:54, 2:63, 3:26, 5:55, 5:58, 5:61, 5:115
ControlNet 3:26
Counter-Mode/CBC-MAC Protocol (CCMP) 3:89
Critical Infrastructure (CI) 1:4, 1:6, 1:24, 1:76, 1:80, 2:5, 4:85, 4:87, 5:58, 5:97, 5:112-113
Cross Site Request Forgery (CSRF) 4:35, 4:39-40
Cross Site Request Forgery (XSRF) 4:35, 4:40
Cross Site Scripting (XSS) 2:44, 4:35, 4:39
CrowdStrike 5:32
Cryptographic Keys 2:46, 2:48-49, 2:52, 3:61
Cyber Risk Preparedness Assessments (CRPA) 5:106
Cyber Security Incident Response Team (CSIRT) 4:84-86
Cylance 5:32
Cryptographic algorithms 3:62

Control Network Attack Surface 2:8
Cellular Backhauls 3:78
Consequence-driven Risk Prioritization 5:86
Culture of Compliance 5:77-78
Cost Benefit Analysis 5:88

D

Data diode 2:26, 3:31, 3:45, 3:49-50, 3:53, 3:57, 4:24
Data Encryption Standard (DES) 3:62-63, 3:65, 3:80
Data Link Connection Identifier (DLCI) 3:15
data link layer 2:61, 2:65, 2:67, 3:10, 3:15
Data Security (PR.DS) 2:57, 5:58, 5:60
DCOM/RPC 3:29, 4:12
decision tree 4:81
Decryption 3:36, 3:63-64, 3:68-69, 3:73, 3:93
Defense-in-Depth 3:46, 3:56, 5:56
DeMilitarized Zone (DMZ) 2:7, 2:26, 2:28-30, 2:33-34, 2:37, 3:55, 4:24, 4:26-27, 4:58
Denial of Service (DoS) 2:42, 2:51, 2:70, 2:85, 3:37, 3:48, 3:93, 3:98-99, 3:101, 4:19, 5:89
Department of Energy (DOE) 5:94
Department of Homeland Security (DHS) 1:22, 1:24, 1:48, 1:69, 1:80, 2:79-80, 4:81-82, 5:103
Derivative Term 1:31
Detection Processes (DE.DP) 5:58, 5:61
DeviceNet 1:51, 2:64, 3:26
df 5:10-11
Diffie-Hellman 3:62-63, 3:68
Digital I/O 1:36
Digital Protective Relay (DPR) 1:38
Digital Signatures 3:31, 3:60, 3:62-63, 3:68-70
diode 2:26, 3:31, 3:45, 3:49-50, 3:53, 3:57, 4:24
Direct Sequence Spread Spectrum (DSSS) 3:85
Disaster Recovery (DR) 1:70, 4:88, 5:88-91, 5:95-96, 5:103

discrete 1:14-15, 1:18, 1:29, 1:36, 1:54, 2:63, 3:24, 3:26, 3:62, 5:61
disk free 5:10-11
Distributed Control System (DCS) 1:28-29, 1:53, 1:56, 1:58, 1:68, 2:31-32, 2:85-86, 4:12, 4:55
Distributed Network Protocol (DNP) 2:43, 2:62, 3:23, 3:27-28, 3:45, 3:50
Distributed Network Protocol (DNP3) 3:27
DNP3 2:43, 2:62, 3:23, 3:27-28, 3:45, 3:50
DNP3v5 2:43
DNS Spoofing 3:16, 3:38
Docker 5:15
Dragonfly 4:19
drivers 1:12, 1:70, 2:57, 4:54-55, 5:7, 5:42, 5:80
DROP 1:56, 1:77, 2:33, 3:6, 3:46, 3:48, 3:50, 3:53, 3:57, 3:80, 3:100, 5:23-24, 5:26, 5:44

Data diodes 3:50

Digital Signatures 3:70

Databases 4:25

Database Security 4:27

DHS-Suggested Patch Decision Tree 4:81

DHS-Recommended Patch Management 4:82

df 5:11

E

E0 3:64, 3:87

EAP/TLS 3:98
Eavesdropping 3:80
ECC 3:60, 3:62
Electrically Erasable Programmable Read Only Memory (EEPROM) 2:2, 2:42, 2:53-54
Electro Magnetic Transmission (EMT) 3:37
Electronic Communications Privacy Act 5:98

Emerson 1:7, 1:22, 2:65, 3:84
Emerson (DeltaV) 1:7, 1:22, 2:65, 3:84
Encapsulating Security Protocol (ESP) 3:12, 4:85
Encryption 2:49, 3:11-12, 3:29-31, 3:36, 3:40, 3:42, 3:53, 3:60-66, 3:68-70, 3:73, 3:78, 3:80, 3:82, 3:89, 3:93, 3:95, 3:98, 3:101, 4:52, 4:69, 5:95
End of Extended Support 4:54
End of Life (EOL) 4:54-57, 4:66
End of Mainstream Support 4:54
End of Sales 4:54
Endgame 5:32
Endress+Hauser 3:84
Energy Management System (EMS) 1:51, 4:55
Energy Sector Control Systems Working Group (ESCSWG) 2:81
Enumeration 2:12, 2:17, 3:39
Environmental Protection Agency (EPA) 2:64

Eradication 5:94, 5:100-101, 5:109
Error (E) 1:24, 1:30-31, 1:47, 1:74, 1:81, 2:5, 2:16, 2:63, 2:67, 2:84, 2:87, 3:17, 3:27, 3:48, 3:64, 3:79, 3:87, 4:11, 4:45, 5:9, 5:48, 5:70, 5:83, 5:96, 5:106
ESET 5:32
EtherCAT 2:62, 2:64, 3:9, 3:23
EtherNet/IP 2:62-63, 3:23, 3:26
European Programme for Critical 1:24 Infrastructure Protection (EPCIP)

European Union Agency for Network and Information Security (ENISA) 1:22, 4:86
Event Log 1:46, 3:52, 4:25, 5:2, 5:38, 5:44, 5:51-52
Event Viewer 5:38-39, 5:44, 5:51
Exponentiation 3:68
Exposure Factor (EF) 5:85
Extensible Authentication Protocol (EAP) 3:89, 3:98

Enforcement Boundary Devices 3:34

Ethernet 3:5

Enterprise Aggregate Historian Architecture 4:26

ENISA 4:86

EtherNet/IP 3:26

F

F-Secure 5:32
Facilities 1:22-23, 1:68, 1:74-76, 4:13, 4:15, 4:25, 5:47, 5:59-60, 5:98, 5:105, 5:107
Factorization 3:68
Factory Acceptance Test (FAT) 1:68-69, 2:79, 2:82
Federal Energy Regulatory Commission (FERC) 1:22, 5:78, 5:85

Field Devices 1:35
File Replication Service (FRS) 5:38
File Transfer Protocol (FTP) 2:18, 2:42, 3:10, 5:15
FIN 3:18
find 2:8, 2:17, 4:5, 5:6, 5:13, 5:17, 5:22

FIPS PUB 180-1 3:66
FIPS PUB 180-2 3:66
FireEye 2:5, 2:91, 5:32
Firewall Builder 5:21
Firmware attacks 2:49
firmware 1:22, 1:33, 1:80, 2:42, 2:44, 2:46, 2:48-50, 2:52, 2:54, 2:57-58, 2:73, 2:92, 3:72, 3:89, 3:99, 4:25, 4:34, 5:101
Fortinet 5:32
Forward Error Correction 3:79
Framework Implementation Tiers 5:64
Front End Processor (FEP) 2:51, 2:71
Front-End Processor (FEP) 2:51, 2:71
Fully Qualified Domain Name (FQDN) 3:15
Function Block Diagram (FBD) 1:33
Functional Safety 2:89
Fuzzing 2:44, 2:63, 3:39, 4:2, 4:42, 4:47-49
Fuzzy Logic 1:32
Firewall Solution (two-firewall vs single firewall) 2:28
Field Bus 2:64

G

Global Navigation Satellite System (GNSS) 2:68-69
Global Positioning System (GPS) 2:68-69, 3:81
GNU Radio 3:97
GoodFET 2:47-48
Google hacking 2:13, 2:16
Googlebot 2:13
Governance (ID.GV) 1:65, 5:1, 5:58-59, 5:77, 5:79
Government Communications Headquarters (GCHQ) 3:69, 3:73
grep 5:13, 5:19
Group Policy Object (GPO) 4:62, 4:64, 4:73, 5:41

GSM 3:64, 3:77-78, 3:95

Group Policy Objects (GPO) 4:64

H

HART
1:33, 1:37, 2:64-65, 3:23, 3:77, 3:83-86, 4:25, 5:8, 5:115

HART Communication Foundation 3:85

HART Communication Foundation (HCF) 3:85

HART-IP 3:23

Hashed Message Authenticity Check (HMAC) 3:27, 3:67, 3:70, 3:89

Havex 4:19-20

HAZard and OPerability (HAZOP) 1:64, 2:89, 2:92

Hazard Operations (HAZOP) 1:64, 2:89, 2:92

High Voltage Protection (HVP) 3:77

Highway Addressable Remote Transducer (HART) 1:37, 2:64-65, 3:23, 3:84-85

historian 1:43-44, 1:46, 1:70, 2:30-31, 2:69, 3:23, 3:29, 3:45, 3:50, 3:60, 4:2, 4:23-28, 4:41, 5:49, 5:87

Hitachi 1:22

HMAC-MD5 3:67

HoneyNet 3:55

honeypot 3:45, 3:55-57

Honeypot 3:45, 3:55-57

Honeywell 1:22, 2:85, 3:86

Honeywell Safety System Firewall 2:85

Host IDentifier (HOST_ID) 3:15

Human Machine Interface (HMI) 1:3, 1:22, 1:28-29, 1:43-45, 1:47, 1:50, 1:56, 1:60-61, 2:8, 2:14, 2:31, 2:43, 2:51, 3:16, 3:23-24, 3:27-29, 3:45, 3:60, 3:77, 4:5-6, 4:25, 4:33-35, 4:39-40, 4:42, 4:45, 4:54, 4:56, 4:67, 5:87

hybrid 1:14, 1:18, 3:9

Hyper Text Transfer Protocol Secure (HTTPS) 2:18, 3:10, 3:15, 3:35, 4:58, 5:24

Hyper-V 4:53

HyperText Transfer Protocol (HTTP) 2:13, 2:18, 3:10, 4:33, 4:37, 4:40, 4:48, 4:53

High Level Attack Surface 2:7

Hazop 2:89

Host-based firewalls 4:67

I

ICS-CERT 1:48, 1:81, 2:12, 2:79, 2:91, 3:29, 4:80, 4:83-84, 5:103

ICS Process Models 1:14

Idaho National Laboratory (INL) 2:80

Identification 2:20, 2:84, 2:89, 3:11, 4:15, 4:57, 4:80, 5:99

IEC 60870 3:27, 3:30-31

IEC 61508 2:88-89

IEC 61511 2:88

IEC 61513 **2:88**

IEC 61850 2:64, 3:9, 3:23, 3:28, 3:31

IEC 61968 3:31

IEC 61970 3:31

IEC 62061 2:88

IEC 62351 3:31

IEC 62425 2:88

IEC 62443 1:19-20, 2:4, 2:26, 2:84, 2:92, 3:44-45, 3:59, 3:76, 3:92, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:57, 5:66, 5:68, 5:73, 5:82, 5:93

IEC 62591 3:84

IEC-62443 2:84, 5:86

IEEE 2:63, 2:65, 3:5, 3:11, 3:15, 3:23, 3:27, 3:69, 3:84-89, 3:98
IEEE 802.15.4 3:84-86
IEEE 802.1X 3:98
Improvements (RC.IM) 4:45, 4:55, 5:58, 5:62-63, 5:89, 5:101
Improvements (RS.IM) 4:45, 4:55, 5:58, 5:62-63, 5:89, 5:101
Incident Handling 3:46, 5:94-99, 5:101-102
Independent System Operator (ISO) 1:38, 1:55, 2:4, 2:41, 2:56, 2:61, 2:77, 2:88, 3:4, 3:30, 3:34, 3:44, 3:59, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:57, 5:66, 5:68, 5:72-75, 5:82, 5:93
Indicators of Compromise (IOC) 4:70, 5:98
Information Protection Processes and Procedures (PR.IP) 5:58, 5:60
Input/Output (I/O) 1:35, 1:56, 2:82, 3:24, 5:35
Instruction List (IL) 1:33, 4:85
Integral Term 1:31
Intelligent Electronic Devices (IED) 1:20, 1:38-39, 1:56, 2:71, 3:24, 3:27-28
Inter-Control Center Communications Protocol (ICCP) 1:55, 3:23, 3:30, 3:50
Interface Identification 3:11
International Data Encryption Algorithm (IDEA) 3:60, 3:62-63

ISASecure 2:84

ITU-defined ISM Bands 3:94

Incident response 5:93

International Electrotechnical Commission (IEC) 1:19-20, 1:33, 2:4, 2:26, 2:41, 2:56, 2:64-65, 2:77, 2:84, 2:88-89, 2:92, 3:4, 3:9, 3:23, 3:27-31, 3:34, 3:44-45, 3:59, 3:76, 3:84-85, 3:92, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:57, 5:66, 5:68, 5:73, 5:82, 5:86, 5:93

International Society of Automation (ISA) 1:12, 1:19-20, 2:4, 2:26, 2:41, 2:56, 2:63, 2:65, 2:77, 2:84, 3:4, 3:14, 3:34, 3:44-45, 3:59, 3:76-77, 3:83, 3:85-86, 3:92, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:57, 5:66, 5:68, 5:73, 5:82, 5:86, 5:93
Internet Assigned Numbers Authority (IANA) 3:13
Internet Control Message Protocol (ICMP) 3:10, 3:12, 3:17, 3:100, 5:24
Internet Protocol (IP) 1:12, 1:51, 1:54, 1:59, 2:8, 2:13-14, 2:17, 2:33, 2:58, 2:62-64, 2:66-67, 2:72, 2:81-82, 3:2, 3:4, 3:6, 3:9-17, 3:19, 3:23-26, 3:32, 3:35-38, 3:50, 3:53, 3:56, 3:79-82, 3:86, 4:6, 4:8, 4:62, 4:69, 4:72, 4:78, 4:88, 5:7, 5:12, 5:23, 5:48, 5:60, 5:95, 5:98
Internet-based Time Service (ITS) 2:70
Interprocess Communication (IPC) 2:59
InterProcess Communication (IPC) 2:59
intractable 3:68
Intrusion Detection System (IDS) 3:16, 3:31, 3:45, 3:47, 3:49, 3:51, 3:56, 3:99, 5:56, 5:97
Invensys 1:7
ipchains 5:21
ipfilter (ipf) 5:21
ipfw 5:21
ipfwadm 5:21
IPSec 2:63, 3:12, 3:46, 3:80, 4:52-53
iptables 5:16, 5:21-25
IPv4 3:10-13, 3:15, 3:17, 5:25-26
IPv6 2:63, 3:10-15, 3:17, 3:19, 3:85-86, 5:24-26
ISA Security Compliance Institute (ISCI) 2:84
ISA-62443 1:12
ISA100 3:77, 3:83, 3:85-86
ISA100.11a 3:77, 3:83, 3:85-86
ISA99 1:20, 2:84

ISO 26262 2:88
ISO 27001 5:57, 5:66, 5:72, 5:74-75
ICS DMZ 2:29
ISACs (information sharing and analysis centers) 4:87
iptables 5:22
I-4 Satellites 3:81

J

jail() 5:15

K

Ka-Band (26.5-40GHz) 3:79-80
Kaspersky Lab 4:18, 5:32
Kerberos 4:46, 4:52-53, 4:71
Key Reinstallation Attack (KRACK) 3:89
Keyspace 3:61
kill 5:13, 5:103
killall 5:13
Ku-Band (12-18GHz) 3:79-80

L

Ladder Diagram (LD) 1:33
Layers of Protection Analysis (LOPA) 1:64, 2:89, 5:86
LDAP 2:30, 4:46, 4:53
Lessons Learned 2:80, 5:62-63, 5:72, 5:102-103, 5:105, 5:108
Level 0 1:19, 1:25, 1:28, 1:53, 1:82, 2:2, 2:26, 2:31-32, 2:36, 2:41-42, 2:52, 2:56, 2:73, 2:77, 2:92, 4:67, 5:18, 5:37, 5:86-87
Level 1 1:18-19, 1:25, 1:28, 1:53, 2:26, 2:32, 4:34, 5:49, 5:86-87

Level 2 1:19, 1:25, 1:28, 1:44, 1:50, 1:53, 2:26, 2:30-31, 2:34, 2:37, 2:85, 3:29, 4:6, 4:33, 4:45, 4:58, 5:86-87
Level 3 1:19, 1:25, 1:44, 1:51, 1:53-54, 1:59, 2:26, 2:29-31, 2:33, 2:36-37, 3:51, 3:57, 4:21, 4:26-27, 4:33, 4:58, 4:89, 5:18, 5:37, 5:8687, 5:90
Level 4 1:19, 1:25, 2:26-27, 2:29, 4:26, 5:18, 5:86-87
Line of Sight (LOS) 3:79
Linux Commands 5:12
Local File Inclusion (LFI) 4:35, 4:41
Local Security Policy 2:33, 4:60, 4:62
locate 3:99, 5:13
Logarithm 3:68

LonWorks 1:51
low-level machine code 1:33
LTE 3:77-78
LXC/LXD 5:15
Lynis 5:16, 5:20, 5:29

Local File Inclusion 4:41

Linux vs Windows file layout 5:9

Linux vs Windows commands 5:12

Linux Containers 5:15

M

Maintenance (PR.MA) 1:9, 1:24, 1:47-48, 1:56, 1:66, 1:68, 1:70, 2:14, 2:42, 2:45-46, 2:57, 2:61, 2:78-80, 2:82, 3:23, 3:29, 3:84-86, 4:33, 4:56-57, 4:80-81, 5:25, 5:38, 5:58, 5:60, 5:75, 5:87-88

malware 1:80, 2:11, 2:14, 2:33, 2:81, 2:91, 3:37, 3:54, 4:11-12, 4:18-21, 4:34, 4:55, 5:3234, 5:100, 5:103
Malwarebytes 5:32
Mandatory Access Control (MAC) 3:5-6, 3:11-12, 3:15-16, 3:35, 3:62, 3:67, 3:78, 3:82, 3:84-86, 3:93, 3:97-98, 5:26, 5:35
Manipulated Variable (MV) 1:30
Masquerading 3:93, 3:98
Maximum Tolerable Downtime (MTD) 5:90
mbtget 3:41-42
McAfee 5:32, 5:48
MD5 3:60, 3:62, 3:66-67, 5:36
Measuring Error Conditions 1:31
Media Access Control (MAC) 3:5-6, 3:11-12, 3:15-16, 3:35, 3:62, 3:67, 3:78, 3:82, 3:84-86, 3:93, 3:97-98, 5:26, 5:35
Merkle 3:69
Message Digest 2 (MD2) 3:66
messages log 5:45-46
Metasploit 1:68, 2:19, 3:98
microkernel 2:59
Microsoft 3:29, 4:2, 4:8, 4:16, 4:46, 4:51-60, 4:6264, 4:66, 4:69, 4:71, 4:74, 4:76, 4:83, 5:57, 5:14-15, 5:32, 5:34-35, 5:38, 5:40, 5:4243, 5:48
Microsoft Management Console (MMC) 4:62
Midcontinent Independent System Operator (MISO) 1:55, 3:30
Mitigation (RS.MI) 2:11, 2:78, 3:99, 4:54, 4:56-57, 4:66, 4:87, 5:58, 5:62, 5:85, 5:90
Mobile Device Management (MDM) 5:5
Modbus 1:51, 2:62-67, 2:71, 2:74, 2:85, 3:2, 3:20, 3:23-24, 3:37, 3:39, 3:41-42, 3:45, 3:47, 3:50

Modbus TCP 2:62, 2:64, 2:66-67, 3:2, 3:20, 3:23-24, 3:41-42
Modbus Mapped to OSI Model 2:66
Modbus TCP mapped to Modbus RTU 2:67
ModbusPal 3:41
monolithic kernel 2:59
Multiplication 3:68
Message Authentication Code (HMAC) 3:67
Malware types 3:11
Microsoft Client OS versions 4:52
Microsoft Server OS versions 4:53
Microsoft Endpoint Configuration Manager 4:59
Modbus TCP 3:24
Mikrokernel OS 2:59

N

National Critical Information Infrastructure Protection Centre (NCIIPC) 1:24
National Electric Sector Cybersecurity Organization Resources (NESCOR) 1:2, 2:1, 3:1, 4:1, 5:1
National Institute of Standards and Technology (NIST) 1:2, 1:22, 1:29, 1:50, 1:53, 1:57, 2:1, 2:4, 2:28, 2:41, 2:56, 2:70, 2:77, 3:1, 3:4, 3:34, 3:44, 3:59, 3:66, 3:76, 3:92, 4:1, 4:4, 4:23, 4:32, 4:51, 4:63, 4:78, 5:1, 5:4, 5:31, 5:54, 5:57-66, 5:68, 5:73, 5:82, 5:93-94
National Security Agency (NSA) 3:69
National Technical Research Organisation (NTRO) 1:24
National Vulnerability Database (NVD) 2:12
Neighbor Discovery Protocol (NDP) 3:15-16, 5:26
Netcat 3:42
netstat 5:13
Network Access Control (NAC) 3:8, 4:52-53
Network IDentifier (NET_ID) 3:15

Network Intrusion Detection System (NIDS) 2:26, 3:24, 3:45, 3:51, 3:57

network layer 2:61, 2:67, 3:10, 3:15, 3:17, 3:35, 3:85
Network Load Balancing (NLB) 4:53
Network Prefix 3:11
Network Time Protocol (NTP) 2:68-70, 3:10
nftables 5:21-22, 5:25-26
NIST CSF 2:4, 2:41, 2:56, 2:77, 3:4, 3:34, 3:44, 3:59, 3:76, 3:92, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:57-64, 5:66, 5:68, 5:82, 5:93
NIST SP 800-53 2:4, 2:41, 2:56, 2:77, 3:4, 3:34, 3:44, 3:59, 3:76, 3:92, 4:4, 4:23, 4:32, 4:51, 4:78, 5:4, 5:31, 5:54, 5:68, 5:82, 5:93
NIST SP800-82 1:50, 1:53, 1:57, 2:28
nmap 2:13, 2:17-18, 2:21
Nmap 2:13, 2:17-18, 2:21
Nmap Scripting Engine (NSE) 2:17

non-routable IP 3:13
North American Electric Reliability Corporation (NERC) 1:22, 5:73, 5:76, 5:80, 5:85, 5:105-106, 5:113

nftables 5:25

NIST CSF 5:57

O

Object Linking and Embedding (OLE) 3:29, 4:19
Object Linking and Embedding for Process Control (OPC) 3:23, 3:29, 3:39, 3:45, 3:50, 4:19-20, 4:24, 4:33
Open DeviceNet Vendors Association (ODVA) 3:26

Open Systems Interconnect (OSI) 1:46, 2:60-61, 2:66-67, 2:85, 3:6, 3:10, 3:26, 3:35, 3:45, 3:84, 5:25

Out of Band (OOB) 3:63, 3:87
Over-The-Air (OTA) 3:85

P

Palo Alto Networks 5:32
Panda Security 5:32
patch decision tree 4:81
payload 2:66-67, 3:12, 3:45, 3:70, 3:73, 3:84, 4:11, 4:15, 4:18-19
PEAP 3:89, 3:98
Pepperl+Fuchs 3:84
Personal Protective Equipment (PPE) 1:64
Phasor Data Concentrator (PDC) 1:38
Phasor Measurement Unit (PMU) 1:38, 2:69
Phishing 2:14, 4:9-10, 4:34
physical layer 2:61, 2:65
POWERLINK 2:62, 2:64
PowerShell 4:2, 4:52-53, 4:66, 4:69, 4:71-73(wmic), 4:75-76, 5:36, 5:40, 5:42-43
Precision Time Protocol (PTP) 2:68-69
Preparation 5:43, 5:94-96, 5:109
presentation layer 2:61, 3:35
Presidential Policy Directive 21 (PPD-21) 1:24
Pretty Good Privacy (PGP) 3:80, 4:84
Process Control System (PCS) 1:18, 1:50
Process Hazard Analysis (PHA) 1:64, 2:89, 2:92
Process ID (PID) 1:30-31, 5:59
process model 1:14, 1:18
Process Variable (PV) 1:30, 1:32
PROFIBUS 1:51, 2:62, 2:64-65, 2:71, 3:25, 4:80
ProfiNet 2:62-65, 3:9, 3:23, 3:25, 4:80
PROFINET 2:62-65, 3:9, 3:23, 3:25, 4:80

Programmable Logic Controller (PLC) 1:3, 1:22, 1:28-29, 1:33-34, 1:40-41, 1:50, 1:53, 1:56, 1:60, 2:32, 2:42-43, 2:50, 2:66-67, 2:74, 2:88, 3:16, 3:24, 3:27-28, 3:45, 4:67, 5:87
Project SHINE 2:18
Proof-Of-Concept (PoC) 2:50
Proportional Term 1:31
Protective Technology (PR.PT) 5:58, 5:60
ps 5:13
Purdue Enterprise Reference Architecture (PERA) 1:19
Purdue Levels 0 and 1 1:27
Purdue Levels 0 and 1 Attacks 2:42
pwd 5:13
Profinet versions 3:25
Password Policies in ICS 4:43
PowerShell 4:71
Policy Hierarchy 5:69
Ports in ICS 3:24
Procedures 5:71
Procurement Language 2:79
Process Engineer 1:21
Purdue Level 0/1 Attack Surfaces 2:42

Q

QNX 2:58-59
Quality of Service (QoS) 2:63, 3:79, 3:85
Quality of Service (QOS) 3:11-12, 3:79, 3:85
Quantitative Risk Analysis (QRA) 1:64, 5:83

R

Radio Frequency (RF) 1:80, 2:48, 3:35, 3:37, 3:77, 3:79-80, 3:82-83, 3:86, 3:93-94, 3:96, 3:99-100
RADIUS 2:69, 3:89, 4:43, 4:46-47, 4:49
RC4 3:60, 3:62, 3:64, 3:89

real-time 1:19, 1:29, 1:33, 1:38, 1:50, 1:55, 2:31, 2:42, 2:58-59, 2:62, 2:64, 3:9, 3:12, 3:18, 3:25, 3:30, 4:24-25, 5:48
Real-Time Operating Systems (RTOS) 2:42, 2:57-59, 2:73, 5:5
Recovery 4:45, 4:88, 5:18-19, 5:48, 5:58-59, 5:6263, 5:76, 5:88-90, 5:94, 5:96, 5:99, 5:101, 5:103, 5:109
Recovery Planning (RC.RP) 5:58, 5:63
Red Hat Enterprise Linux (RHEL) 5:5
REJECT 2:69, 3:56, 5:23
Remote Access Trojan (RAT) 1:77
Remote Code Execution (RCE) 2:51, 4:66
Remote Desktop Protocol (RDP) 2:33, 3:35, 4:52, 4:67
Remote File Inclusion (RFI) 4:35, 4:41
Remote Procedure Control (RPC) 3:29, 4:12, 4:17, 4:33
Remote Terminal Unit (RTU) 1:56, 2:32, 2:64, 2:66-68, 2:74, 3:24, 3:27-28, 5:87
Requested Packet Interval Rate (RPI) 2:63, 3:26
Response Planning (RS.RP) 5:58, 5:62, 5:76
RF Mesh 3:82-83
Risk Assessment (ID.RA) 1:20, 1:75, 2:14, 5:58-59, 5:65, 5:74, 5:83-84, 5:86, 5:107
Risk Management Strategy (ID.RM) 5:58-59
Rockwell Automation 1:7, 1:22
RS-232 1:37, 1:56, 2:42
RS-422 1:37
RS-485 1:37, 1:56
RSA 1:67, 2:14, 2:17, 2:45, 2:62, 2:67, 2:80, 2:91, 3:37, 3:45-47, 3:57, 3:60, 3:62, 3:67, 3:78-79, 3:81, 3:94, 3:97, 5:45, 5:55
runlevel 5:18-19
RTOS Kernel 2:59
RF Mesh Networks 3:83
RTU 1:56

S

Safety Instrumented Function (SIF) 2:87, 5:87
Safety Instrumented System (SIS) 1:28, 2:85-89, 2:91-92, 3:45, 3:50, 5:87

Safety Integrity Levels (SILs) 2:89
Samurai Security Testing Framework for Utilities (SamuraiSTFU) 1:2, 2:1, 3:1, 4:1, 5:1
Samurai Web Testing Framework (SamuraiWTF) 1:2, 2:1, 3:1, 4:1, 5:1
SCALANCE S602 2:85
Schneider Electric 1:22, 1:49, 2:91
Schneider Electric (Citect) 1:22, 1:49, 2:91
secedit.exe 4:62
Secure Hash Algorithm (SHA) 3:60, 3:62, 3:66-67, 3:89
Secure Hash Standard (SHS) 3:66
Secure SHell (SSH) 2:18, 2:63, 3:10, 3:35, 3:80, 4:67, 5:14-15, 5:24, 5:48
Secure Simple Pairing (SSP) 3:87
Security Continuous Monitoring (DE.CM) 5:58, 5:61
Security Templates 4:62-63
SentinelOne 5:32
Sequential Function Chart (SFC) 1:33
session layer 2:61, 3:10, 3:26, 3:35
SetPoint (SP) 1:30, 1:32-33, 1:50, 1:53, 1:57, 2:4, 2:28, 2:41, 2:56-57, 2:65, 2:77, 3:4, 3:30, 3:34, 3:44, 3:59, 3:76, 3:92, 4:4, 4:23, 4:25, 4:32, 4:51, 4:71, 4:78, 5:4, 5:31, 5:54, 5:68, 5:82, 5:93
SHA-1 3:66-67
SHA-256 3:66
SHA-384 3:66
SHA-512 3:66-67
Shodan 2:13, 2:18-19, 2:21-22
Siemens 1:22, 1:29, 2:19, 2:85, 2:90, 3:84, 4:13, 4:16, 4:80
SIMATIC 1:29, 4:17
Single Loss Expectancy (SLE) 5:85
Site Acceptance Test (SAT) 1:68-69, 2:79, 2:82, 4:74, 5:37
Small Business Server (SBS) 4:53

SMART goals 5:72
Smart Grid Interoperability Panel (SGIP) 1:2, 2:1, 3:1, 4:1, 5:1

SOAP 1:51, 3:29
Softing 3:84
Software Restriction Policies (SRP) 4:62, 5:34-35
Solaris 5:5, 5:9-10, 5:15, 5:20-21
Sophos 5:32
SQL Injection (SQLi) 3:39, 4:2, 4:27, 4:29-30, 4:35, 4:38
SQL Injection (SQLI) 3:39, 4:2, 4:27, 4:29-30, 4:35, 4:38
Stratum 2:69-70
strings 2:16, 2:53, 3:11, 3:39, 5:48
Structured Query Language (SQL) 1:46, 3:39, 4:2, 4:27, 4:29-30, 4:35, 4:38
Structured Text (ST) 1:33
Stuxnet 1:41, 1:49, 2:7, 4:13-18, 4:20, 4:40, 4:80
Subnet ID 3:11
sudo 2:17, 3:15, 5:13, 5:18-19, 5:23, 5:25
Supply Chain Risk Management (ID.SC) 5:59
Symantec 4:13, 4:15, 5:14, 5:32
symmetric algorithm 3:62, 3:70
SYN 3:18
SYN-ACK 3:18
syslog 3:10, 3:49, 5:18, 5:45-48, 5:50
System On a Chip (SOC) 2:27, 2:54, 4:53, 5:49, 5:95
System Robustness Testing (SRT) 2:84
System V 1:70, 2:84, 2:90, 3:29, 4:79, 5:9, 5:17-18, 5:20
SIL 2:89

Symmetric key 3:62

SATCOM 3:79

Stuxnet 4:15

systemd 5:19

System V 5:18

syslog 5:46

SMART objectives 5:72

Security Policies Windows 4:62

Security Templates Windows 4:62

T

Table Top eXercise (TTX) 5:105-107, 5:111
tap point 3:49
TC 57 3:31
Telecommunications Industry Association (TIA) 1:37, 2:65-66
telinit 5:18
Telnet 2:18, 2:42, 3:10
Telvent 1:49
Temporal Key Integrity Protocol (TKIP) 3:89
TIA-232 1:37
TIA-422 1:37
TIA-485 1:37, 2:65-66
Time Division Multiple Access (TDMA) 3:79, 3:85
Toshiba 2:64
Total Phase 2:47-48
Transmission Control Protocol (TCP) 1:54, 1:59, 2:8, 2:43, 2:58, 2:62-64, 2:66-67, 2:72, 2:81-82, 3:2, 3:4, 3:9-10, 3:12, 3:17-20, 3:23-25, 3:32, 3:35-38, 3:41-42, 3:47, 3:49-50, 3:53, 3:80, 3:82, 4:6, 4:8, 4:69, 4:72, 5:7, 5:13, 5:23-24
transport layer 2:61-62, 2:67, 3:10, 3:17-18, 3:30, 3:35, 5:50
Trend Micro 5:32

Triple DES 3:63
TrustedBSD 5:35
TTLS 3:98
TCP Ports in ICS 3:23
Two Factor Authentication 4:45
Transparent Firewall in ICS 2:85

U

Unidirectional Gateway 3:50
UNIX commands 5:12-13
User Datagram Protocol (UDP) 2:64, 2:82, 3:10, 3:12-13, 3:17-18, 3:2324, 3:26, 3:35, 3:47, 3:49, 3:53, 4:69, 5:13, 5:24, 5:47, 5:50

V

VBScript 4:73
Velocio 1:40-41, 1:60, 2:74
Very Small Aperture Terminal (VSAT) 3:77, 3:79-80
Virtual LAN (VLAN) 3:7-8, 3:19, 3:78
Virtual Local Area Networks (VLANs) 2:43, 2:63, 3:7-8, 3:37
VMware 4:53, 4:71
Vulnerabilities 2:5, 2:9, 2:11-12, 2:16, 2:20-21, 2:42-44, 2:46, 2:49, 2:51, 2:54, 2:58, 2:63, 2:80, 2:82-83, 2:85, 3:29, 3:39, 3:46, 3:55, 3:57, 3:78, 3:80, 3:93, 3:99, 4:8, 4:16, 4:20, 4:27, 4:29-30, 4:35, 4:38-39, 4:41, 4:44, 4:48, 4:54-55, 4:57, 4:79-80, 4:84, 5:83-84, 5:89, 5:91, 5:100
Vulnerability Identification Testing (VIT) 2:84

VxWorks 2:58

VSAT 3:79

Virtualization in ICS 1:70

W

Weak Session Management 4:35, 4:37
Wi-Fi Protected Access (WPA) 3:89, 3:97-98, 4:52
WiMax 3:79
WinCC 4:17
Windows Embedded Compact 2:58
Windows Event Viewer 5:39
Windows Firewall 4:60, 4:68-69, 5:22
Windows Server Essentials 4:53
Windows Server Update Services (WSUS) 4:58-59
Windows Storage Server 4:53
Wired Equivalent Privacy (WEP) 3:89
Wireless Industrial Technology Konsortium (WiTECK) 3:84
WirelessHART 2:64, 3:77, 3:83-86
Wireshark 2:75, 3:21, 3:32, 3:100
WirlessHART 3:84
Wifi Security 3:89
Wireless Backdoors 3:100
wf.msc (Windows Firewall) 4:68
Windows Server Update Services 4:58
Windows Logs 5:38
wmic.exe 4:72
WPA2 3:89
Wi-Fi security 3:89
WPA 3:89
Windows Operating System verions 4:53

X

x86 5:9
XML 1:51, 3:23, 3:29, 5:44
xxd 2:53

Z

Zed Attack Proxy (ZAP) 4:48
ZigBee 3:14, 3:77, 3:83-84, 3:86, 3:94-95
ZigBee 3:86